IN3061 Tiny Data Science Project - Exploring Disagreement Between Star Ratings and Textual Sentiment in Online Reviews

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1. Introduction (227 words)

Online review platforms summarise user opinion primarily through numeric star ratings, these ratings are compact, easily comparable, and well suited to aggregation at scale. However, star ratings inevitably compress complex user experiences into a single value. In contrast, free-text reviews allow users to express nuance, mixed sentiment, emphasis, and contextual dissatisfaction that may not be fully reflected in a numeric score.

While a large body of existing work uses such data to predict business outcomes such as restaurant success or closure, this analysis adopts a different perspective. The motivation came from a recurring pattern observed in practice: users sometimes describe a restaurant positively while assigning a low star rating, or vice versa, a high rating while expressing criticism in text. These apparently inconsistent reviews are often treated as noise or modelling error, yet they may instead reflect ambivalence, contextual judgement, or the limitations of numeric summaries.

This project therefore investigates the relationship between numeric star ratings and sentiment expressed in review text, with particular emphasis on cases where these two signals disagree. Rather than treating such disagreement as an error, it is considered a potentially informative feature of how users communicate evaluation in practice. To support this, the analysis adopts an exploratory and explanatory approach, focusing on data wrangling, interpretable feature engineering from text, descriptive analysis, and the use of simple models for explanation rather than optimisation.

2. Analytical Questions and Data (255 words)

2.1 Analytical Questions

This analysis focuses on a single phenomenon: the disagreement between numeric star ratings and sentiment expressed in review text. Such disagreement is not necessarily a sign of dissatisfaction or error, but may reflect the different ways users express their opinions through star ratings and written reviews. For this reason, the analysis approaches the phenomenon from three distinct but related angles: how frequently disagreement occurs, how it manifests directionally, and which observable characteristics of review text are associated with stronger mismatches. These considerations motivate the following analytical questions:

1. How frequently do star ratings and text derived sentiment signals disagree within Yelp reviews?
2. Which observable textual characteristics are associated with larger rating-text mismatches?
3. Is disagreement symmetric, or do certain types of mismatch occur more frequently?

These questions were defined prior to detailed analysis and were refined iteratively as exploratory analysis revealed structure within the data, consistent with an exploratory data science approach.

2.2 Data Description

The data used in this project come from the Yelp Open Dataset and only review and business dataset will be choosen for this project, both provided in JSON format, each record represents a single user review and includes both structured metadata and unstructured free text.

The primary variables used are:

- stars: ordinal rating from 1 to 5
- text: free-text review content
- useful: number of usefulness votes
- date: timestamp of the review

Due to the size of the full dataset, the analysis focuses on a controlled subset of approximately 500,000 reviews, loaded incrementally to ensure memory-safe processing.

3. Analysis (1017 words)

3.1 Data Wrangling and Preparation

The review data are stored in JSON format and were too large to load safely into memory in a single operation, so reviews were loaded incrementally using chunked reading, enabling early inspection of data quality and supported iterative development.

Reviews with missing or empty text and missing star ratings were removed, while star ratings were cast to numeric types, date fields converted to datetime objects, and count variables such as useful validated to ensure data credibility.

The business dataset was left joined to the review data to retain business-level identifiers and contextual information while preserving all reviews, even where metadata were incomplete. Although business variables were not used directly in the final analysis, the join supported exploratory checks and informed subsequent feature engineering decisions.

3.2 Feature Engineering

3.2.1 Text-Derived Sentiment

Textual sentiment was extracted using VADER (Valence Aware Dictionary and sEntiment Reasoner), a lexicon-based sentiment analysis method designed for short, informal text such as online reviews. For each review, VADER produces a compound sentiment score in the range $[-1,1]$, representing the overall polarity of the text from strongly negative to strongly positive.

This lexicon-based, single numeric representation was chosen because it enables direct comparison between sentiment expressed in review text and the corresponding numeric star rating, directly supporting the objective of identifying agreement and disagreement between these two evaluative signals.

3.2.2 Additional Textual Features

Several additional interpretable features were derived from the review text, including word count, punctuation usage (exclamation and question marks), and the proportion of capitalised characters, these features capture expressiveness and emphasis while avoiding complex linguistic assumptions.

3.2.3 Defining Disagreement

Disagreement between star ratings and text sentiment was operationalised in two complementary ways.

First, categorical mismatch groups were defined:

- high star ratings (≥ 4) with negative sentiment
- low star ratings (≤ 2) with positive sentiment
- broadly aligned cases

Table 1 summarises the counts and proportions of reviews in each mismatch category, demonstrating that disagreement constitutes a non-trivial fraction of the dataset. To support subsequent explanatory modelling, a binary indicator was also derived to flag cases of strong disagreement between star ratings and text derived sentiment.

mismatch_type	*n_words*	*n_exclaim*	*n_question*	*caps_ratio*	*useful*	*sent_compound*	*stars*
aligned_or_mild	98.203980	1.338191	0.138204	0.006310	0.977151	0.674324	4.047708
neg_text_high_stars	81.334827	1.036812	0.165813	0.008475	0.932458	-0.462348	4.545774
pos_text_low_stars	148.560510	0.899635	0.400660	0.006887	1.510235	0.682187	1.511425

Table 1: Summary statistics of review features by mismatch category.
Mean values of selected textual and metadata features across alignment and disagreement categories

3.3 Exploratory Analysis and Visualisation

3.3.1 Sentiment Across Star Ratings

Figure 1 shows the distribution of VADER compound sentiment scores for each star rating level. While average sentiment increases monotonically with star rating, substantial overlap is observed across rating categories. This indicates that reviews with identical star ratings can express markedly different textual sentiment.

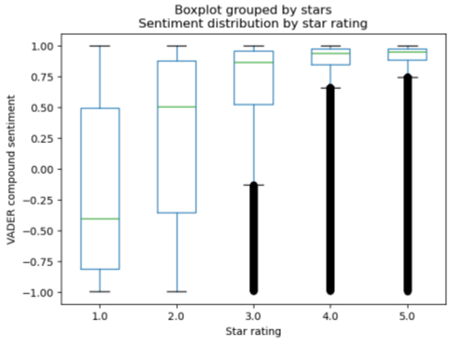

Figure 1: Distribution of sentiment scores by star rating.
Boxplots showing the distribution of VADER compound sentiment scores across star rating levels.

3.3.2 Frequency and Direction of Disagreement

Figure 2 displays the frequency of reviews in each mismatch category. A clear asymmetry is evident, reviews with positive star ratings but negative sentiment occur more frequently than reviews with negative ratings and positive sentiment. This suggests that users may decouple numeric ratings from textual critique when expressing nuanced evaluations.

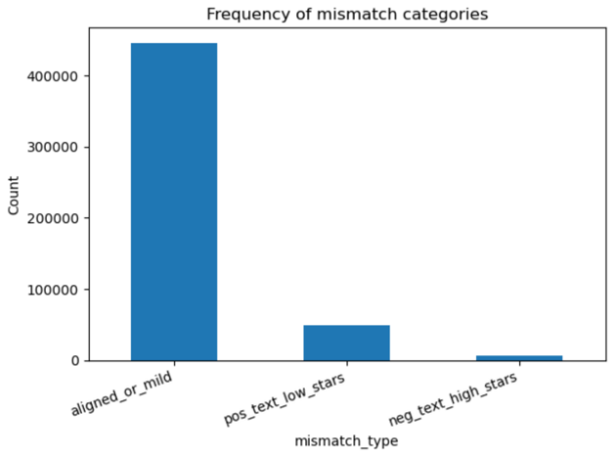

Figure 2: Frequency of mismatch categories.
Bar chart showing counts of aligned reviews and the two directional disagreement categories.

3.3.3 Relationship with Review Length and Usefulness

To examine whether disagreement is associated with expressiveness, review length was analysed.

Figure 3 shows disagreement magnitude across review length bins, longer reviews tend to exhibit stronger disagreement, consistent with the idea that longer text allows for mixed sentiment and contextual explanation that cannot be captured by a single rating.

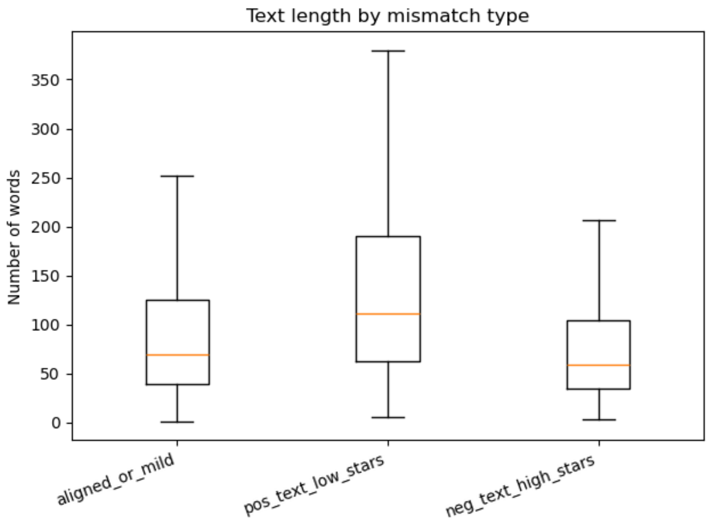

Figure 3: Disagreement magnitude by review length.
Boxplots of disagreement scores across bins of review word count.

Usefulness votes were also examined. Figure 4 shows the relationship between usefulness votes and disagreement, reviews which are marked as more useful often exhibit greater disagreement, suggesting that reviews valued by other users may contain nuanced or balanced evaluations.

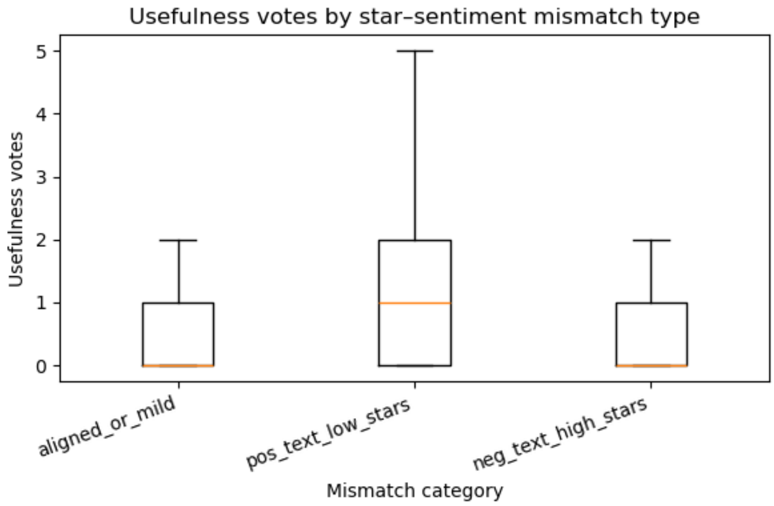

Figure 4: Disagreement versus usefulness votes.
Summary plot showing disagreement magnitude across usefulness vote levels

3.4 Interpretable Modelling for Explanation

To investigate which features are associated with strong disagreement, a logistic regression model was fitted due to its simplity and efficiency while still providing very good interpretability for this topic.

Table 2 reports the standardised model coefficients, sentiment polarity emerges as the strongest driver of disagreement classification, followed by review length. Expressive features such as exclamation marks and capitalisation contribute modestly, while usefulness votes show a positive association with disagreement.

	precision	recall	f1-score	support
0	0.933	0.717	0.811	111250
1	0.203	0.582	0.301	13750
macro avg	0.568	0.650	0.556	125000
weighted avg	0.853	0.702	0.755	125000

Table 2: Classification performance metrics for disagreement model

Standardised coefficients indicate the direction and relative strength of association between features and strong disagreement (accuracy = 0.702)

Model coefficients were inspected to assess feature associations, with performance metrics reported for completeness, showing that disagreement reflects combined sentiment and expressive detail rather than sentiment alone.

Figure 5 shows the confusion matrix for the disagreement classifier. The matrix was used to inspect class-level behaviour rather than to optimise accuracy, reinforcing the explanatory focus of the model.

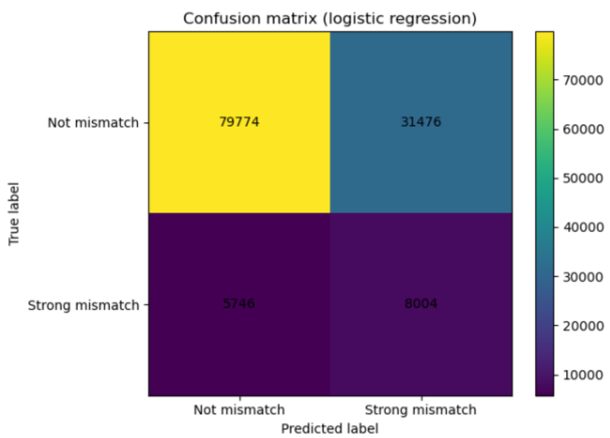

Figure 5: Confusion matrix for disagreement classification.

Confusion matrix showing predicted versus actual disagreement classes

From the confusion matrix, the model identifies 8,004 true positives and 5,746 false negatives for the disagreement class, yielding a recall of 0.582 (8004 / (8004 + 5746)). However, it also produces 31,476 false positives, resulting in a low precision of 0.203 (8004 / (8004 + 31476)). This asymmetric behaviour is consistent with the associations reported in Table 2, while sentiment polarity and review length are strongly linked to disagreement, these features are not exclusive to disagreement cases. Consequently, the model is sensitive to disagreement-related characteristics, leading to high recall, but also flags many reviews that exhibit similar expressive patterns without meeting the strict binary definition of strong disagreement. This supports the interpretation-focused use of the model and reinforces earlier exploratory findings that disagreement is diffuse and not sharply separable.

4. Findings and Conclusions (522 words)

This section summarises the key findings of the analysis, reflects on the extent to which they address the research questions, and evaluates their usefulness, limitations, and implications for further investigation.

4.1 Research Questions

The analysis shows that disagreement between numeric star ratings and text-derived sentiment is common, directly answering the first research question. This demonstrates that ratings and review text often encode different evaluative signals, rather than representing interchangeable measures of user opinion. The presence of substantial overlap in sentiment across rating levels confirms that disagreement is a systematic feature of the data rather than an anomaly.

The second research question is addressed through the clear asymmetry observed in disagreement patterns. Reviews with positive star ratings but negative textual sentiment occur more frequently than the reverse, indicating that users often rely on textual reviews to qualify, contextualise, or moderate otherwise favourable ratings. Suggesting that star ratings may function as coarse summaries, while review text is used to express critique, nuance, and justification.

The third research question is addressed by examining review characteristics associated with disagreement. Longer and more expressive reviews exhibit greater mismatch between ratings and sentiment, and reviews perceived as useful by other users tend to display higher levels of disagreement. These findings indicate that disagreement is linked to detailed and nuanced user expression rather than noise, and that such disagreement may add informational value for readers rather than reducing clarity.

Taken together, the findings show that disagreement reflects structured and meaningful variation in how users communicate evaluations, while numeric star ratings compress judgement into a single value, free-text reviews preserve ambivalence, contextual explanation, and mixed sentiment. From a practical perspective, this suggests that relying solely on star ratings may obscure important aspects of user experience, and that incorporating textual analysis could support more informed decision-making for platforms, businesses, and consumers.

4.3 Conclusion and Future Direction

To summarise, the suitability of the data and analytical approach for answering the research questions is generally strong. The large-scale review dataset provides sufficient variability to observe disagreement patterns, while interpretable feature engineering and simple explanatory modelling allow findings to be traced directly back to observable characteristics of the data.

However, several limitations remain. Lexicon-based sentiment analysis cannot fully capture context, sarcasm, or domain-specific language, and the binary definition of strong disagreement necessarily simplifies a more continuous phenomenon. In addition, the analysis focuses on associations rather than causal relationships.

Future work could address these limitations by exploring alternative sentiment representations, incorporating temporal or business-level effects, or examining how disagreement influences downstream outcomes such as consumer choice or business reputation. Moreover, the analysis was constrained by the scale of the raw JSON data, which necessitated working with a subset of the available files rather than the full corpus. While this approach enabled efficient processing and clear exploratory analysis, loading and analysing the complete set of JSON files could provide a more comprehensive view of disagreement patterns and improve the robustness of the findings (e.g. checkin.json, tip.json, user.json). Nonetheless, this project demonstrates that meaningful and actionable insight can be derived from large, real-world text datasets using transparent and interpretable analytical methods aligned with clearly defined research questions.

Reference

1. Yelp (n.d.) Yelp Open Dataset. Available at: https://business.yelp.com/data/resources/open-dataset/ (Accessed: 19 December 2025).

Appendix

Generative AI (ChatGPT) was used in a limited and critical way during the early planning stage of this coursework.

It was used to:

- Brainstorm possible project ideas aligned with the module content
- Refine and clarify potential research questions